

## Part 3 Optimization Exercise V2 - Start from PCB Antenna Designer Session

In this exercise, you start from a saved PCB Antenna Designer session that already contains the H-notch patch geometry and design variables. We are going to use the **Optimize** workflow available in the app.

The session file for this exercise is: HNotchPatchSession.mat

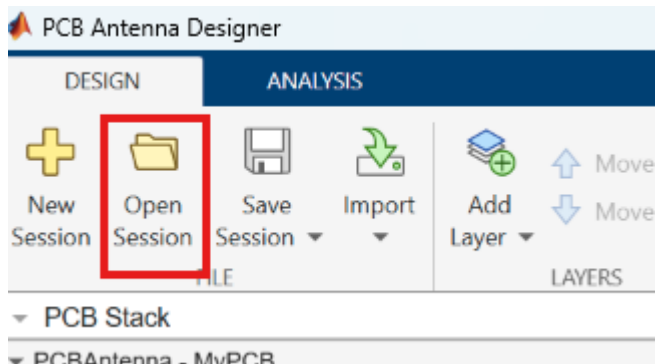
### Open PCB Antenna Designer

Run this command to open the app.

```
pcbAntennaDesigner
```

### Load the Saved Session

In PCB Antenna Designer, use **Open Session** and select HNotchPatchSession/HNotchPatchSession.mat



After the session opens, verify these items before optimization:

- The H-notch patch geometry appears in the 3-D view.
- The **Design Variables** table is populated.
- Patch and notch dimensions are linked to design variables, not plain numeric constants.
- The feed center is  $[-3, -3]$  mm and the feed connects the H-notch layer to the ground plane.

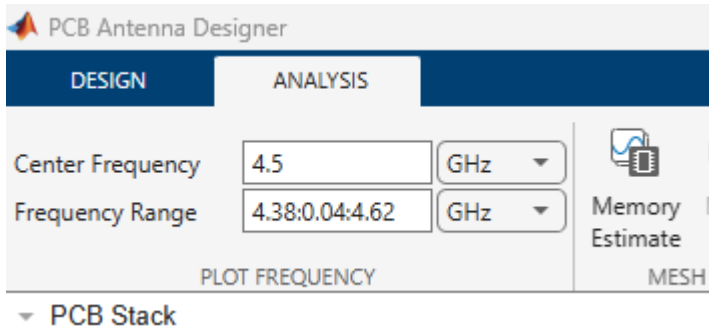
### Set Baseline Analysis Frequencies

Before optimization, go to the **Analysis** tab and set these values so baseline plots use the same frequencies as the optimization.

- **Center Frequency:** 4.5 GHz

- **Frequency Range:** 4.38:0.04:4.62 GHz

```
centerFrequencyGHz = 4.5  
frequencyRangeGHz = 4.38:0.04:4.62
```



## Run Baseline Analysis

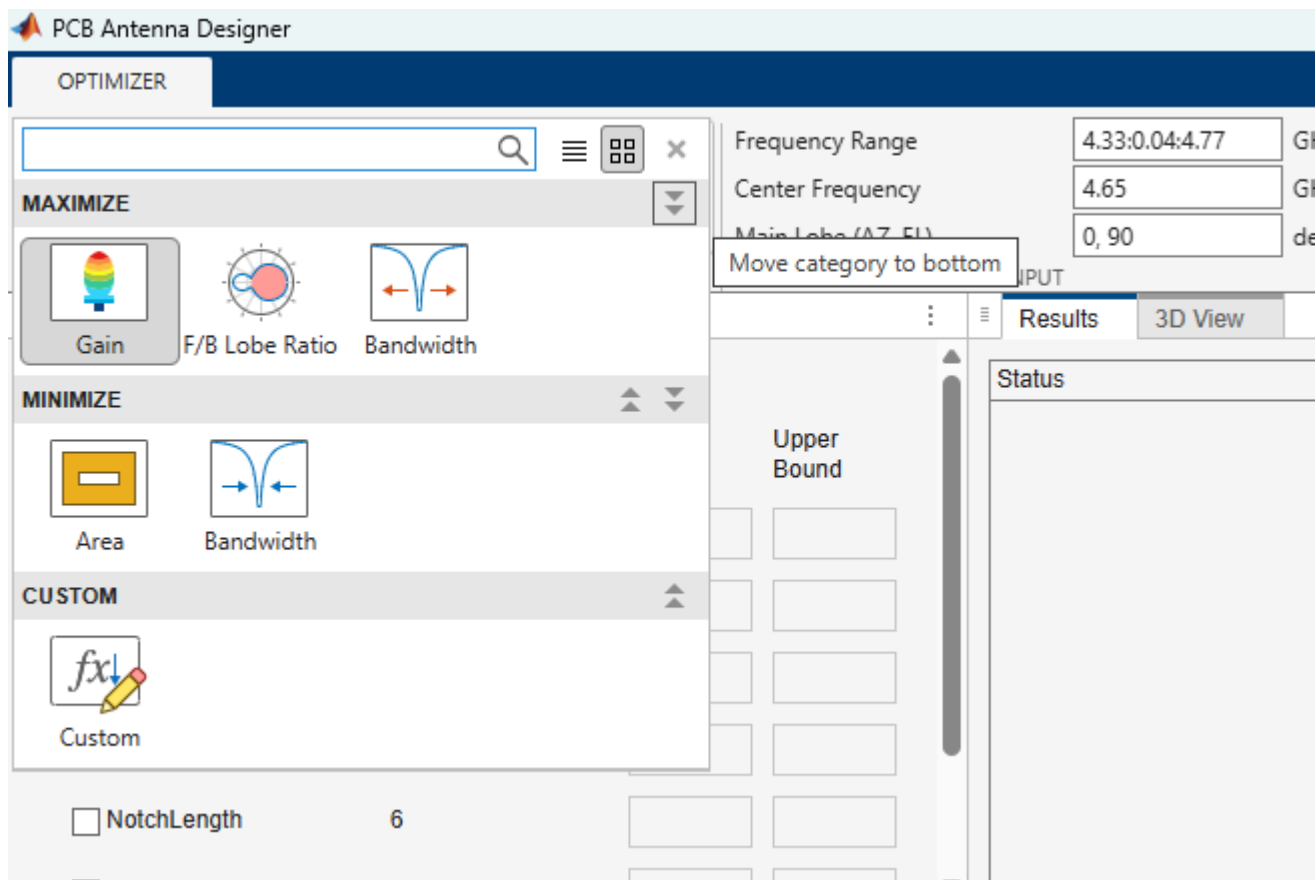
In the **Analysis** tab, run these plots before optimization:

- **Impedance**
- **S Parameter**
- **3D Pattern**

These baseline plots give you a before/after comparison after the optimizer updates the design variables.

## Open the Optimization Workflow

Click **Optimize** in the app toolstrip. In the **OBJECTIVE FUNCTION** gallery, choose **Maximize Gain**.



Keep the main lobe direction at  $[0 \ 90]$  unless the instructor asks you to optimize a different pattern direction.

```
mainLobeDirection = [0 90]
```

## Select Design Variables

On the optimizer **Design Variables** tab, check only the four H-notch patch dimensions shown below. Enter the lower and upper bounds in millimeters.

| Design Variable | Initial Value | Lower Bound | Upper Bound |
|-----------------|---------------|-------------|-------------|
| Length          | 20            | 19          | 24          |
| Width           | 20            | 16          | 21          |
| NotchLength     | 6             | 5.4         | 6.6         |
| NotchWidth      | 6             | 5.4         | 6.6         |

| OBJECTIVE FUNCTION                              |    | INP |     |
|-------------------------------------------------|----|-----|-----|
| ▼ Design Variables                              |    |     |     |
| <input type="checkbox"/> BoardLength            | 25 |     |     |
| <input type="checkbox"/> BoardWidth             | 25 |     |     |
| <input checked="" type="checkbox"/> Length      | 20 | 19  | 24  |
| <input checked="" type="checkbox"/> Width       | 20 | 16  | 21  |
| <input checked="" type="checkbox"/> NotchLength | 6  | 5.4 | 6.6 |
| <input checked="" type="checkbox"/> NotchWidth  | 6  | 5.4 | 6.6 |
| Apply                                           |    |     |     |
| ▼ Geometric Constraints                         |    |     |     |
| ▼ Linear Constraints                            |    |     |     |

Leave board dimensions, feed location, ground plane dimensions, and center coordinates unchecked for this exercise.

```
optimizationVariableNames = ["Length", "Width", "NotchLength", "NotchWidth"];
optimizationBoundsMillimeters = [19 16 5.4 5.4; 24 21 6.6 6.6]
```

## Add the S11 Constraint

On the optimizer **Constraints** tab, configure the impedance-bandwidth constraint.

- **Constraint Function:** S11 (dB)
- **Sign:** <
- **Value:** -8
- **Weight:** 100

```
constraintFunction = "S11 (dB)";
constraintSign = "<";
constraintValue = -8
constraintWeight = 100
```

| % Weight | Constraint Function | Sign | Value | Add | Rem... |
|----------|---------------------|------|-------|-----|--------|
| 100      | S11 (dB)            | N... |       | +   | -      |

None  
Area (m<sup>2</sup>)  
Volume (m<sup>3</sup>)  
S11 (dB)  
Gain (dBi)  
F/B Lobe Ratio (dB)

Apply

Click **Apply**

## Choose Optimization Options

In the optimizer settings area, use these options:

- **Optimizer:** SADEA
- **Iterations:** 100
- **Parallel Computing:** leave unchecked unless the instructor asks you to use a local parallel pool

```
optimizerName = "SADEA";
iterations = 100
```

## Apply and Run

Click **Apply** after entering the objective, design variables, and constraint. Then click the green **Run** button to start SADEA optimization.

During optimization, the app first builds a surrogate model from sampled designs. After that, it runs optimizer iterations and updates the convergence plot.

## Accept the Optimized Design

When optimization completes, click **Accept** to return to the analysis workflow with the optimized dimensions applied to the antenna.

Record the optimized values for Length, Width, NotchLength, and NotchWidth in your notes.

## Validate the Optimized Antenna

In the **Analysis** tab, rerun **Impedance**, **S Parameter**, and **3D Pattern**. Compare the optimized result with the baseline result.

- Does the optimized antenna satisfy  $S_{11} < -8$  dB over any part of 4:0.1:5 GHz?
- Did the peak gain or selected main-lobe gain improve?
- Which design variable changed the most?

## Export the Result

Export the optimized antenna from PCB Antenna Designer to the MATLAB workspace. Use a clear variable name such as hNotchAntOptimized, then save the antenna by itself so the exported MAT-file can also be imported by PCB Antenna Designer later.

```
if exist("hNotchAntOptimized","var")
    save("HNotchPatchUnitElement_Optimized.mat","hNotchAntOptimized")
else
    disp("Export the optimized antenna from PCB Antenna Designer as  
hNotchAntOptimized, then rerun this section.")
end
```

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